

BLOCKS FOR RELIABLE HARDNESS TEST

Table 1 Specification of Standard blocks

New Products: HRB Dual(HRBS+W)d-HRB90,82,72,62,32

YSTL

Assortment	Hardness value	Tolerance	Calibration number(n)	Variation (R=Max.-Min)	Materials (JIS notation)	Dimension (mm)	Finished surface	Standard based
HMV (1, 0.1)	1600	±10%	4 (2x2)	2% (HV1)	Si ₃ N ₄	□10×5	□	JIS B 7735
HMV (1, 0.1, 0.01)	900, 800, 700, 600, 500	±15	6 (3x2)	5% (HV0.1)	SK85	φ25×6	□	JIS B 7735
"	400, 300, 200 (Ble Copper)	±15	"	"	C1720P	φ25×6 (2)	□	"
"	100 (C2600P), 40 (C1020P)	±10	"	"	"	φ25×5 ※	□	"
HMV (0.1, 0.01, 0.001)	30 (Au)	±10	"	"	Au	φ25×5 (0.8)	□	"
UMV (0.01, 0.002)	900, 700 (Berkovich 9.8mN tested)	±20%	6 (3x2)	10% (HV0.01)	SK85	φ25×6	■	JIS B 7735
" (0.01, 0.002)	500, 200 (")	"	"	"	SK85, C1720P	φ25×6 (2)	■	"
★ Nano indentation Hardness Blocks			APPROX 430		Single Crystal	φ25×6	■	JIS B 7735
HN-W (HV0.01, 0.001 Berkovich 9.8mN tested)			HV0.001		Tungsten	(W: φ9×6)		
HV (30, 1)	1000 (SK120), 900, 800, 700	±15	10 (5x2) / HV30, 10	1.5%	SKS3	φ64×15	○	JIS B 7735
HV (10, 1)	600, 500, 400, 300, 200 (SK85), 150 (S45C)	±10	6 (3x2) / HV1	"	"	"	○	"
"	100 (C2600P), 40 (C1020P)	"	"	(150HV and below 2.2%)	"	φ64×10	○	"
HS	100 (SK120), 95, 90, 80, 70, 60, 50, 40, 30	±2	HV10 (5x2)	VHS ≤ 1.5 (70HS and below 1.2)	SK85	φ64×15	○	JIS B 7731
"	20 (S20C), 7 (C1020P φ64×10)	"	HS10 (5x2)	ΔHS (HS-VHS) ≤ 0.5	"	"	○	"
HL	HLE (Dia) 850, 800, 700, 600, 500	±15	(HV Calibration)	"	SK85	φ115×33	○	JIS B 7731
"	HLD (WC) 880, 830, 730, 630, 520	"	"	"	"	"	○	Related
HR C	70 (SK120), 67, 64, 62, 60	±1	10 (5x2)	0.2	SKS3	φ64×15	○	JIS B 7730
"	57, 55, 50, 45, 40, 35, 30, 25, 20, 10	"	"	(40HRC and below 0.3)	SK85	"	○	"
HR A	87, 85, 83, 81, 78, 75, 71, 65, 56	"	"	0.3	Same as HRC	"	○	"
HR30N	83, 81, 78, 73, 67, 60, 55, 50, 41	"	"	0.6	"	"	○	"
HR15N (45N)	92, 90, 87, 85, 80, 75 (43) (23)	"	"	"	"	"	○	"
HRB S	100, 95, 90	±2	10 (5x2)	0.8	SK85	φ64×10	△	JIS B 7730
"	82, 72, 62, 52, 42, 32	"	"	(50HRB and below 1.0)	C2600P	"	△	"
HR30T S	78, 72, 62, 52, 42, 38, 32	"	"	1.0	Same as HRB	"	○	"
HR15T S	87, 82, 78	"	"	"	"	"	○	"
HR (E·M·L·R·F·S)	HRE90 HRM107 HRL118 HRR123 HRF90	"	"	"	(100HV)	"	○	JIS K 7202
"	HRM67 HRL92 HRR105 HRS90	"	"	"	(40HV)	"	○	"
HBW (10/3000)	600, 550, 500, 450, 400, 350	±15	6 (3x2)	1.5%	SK85	φ115×18	●	JIS B 7736
"	300, 250, 229 (10mm), 200, 180	"	"	"	"	"	●	"
"	HBW (10/3000) 150, HBW (10/500) 125	"	"	2.5%	S45C	"	●	"
"	HBW (10/500) 100	"	"	3%	S10C	"	●	"

Finished test surface: ●Fine grinding, △plate lapping, ○Buffing, □Super finish, ■Super finish(fine), ※φ25×6
New Test Blocks (N. T. B.)※(for Spot Anvil)◇Export only

HRC	67, 64, 62, 60	±1	6 (3×2)	φ50.8×6.4 (φ2"×1/4")	(ASTM E-18)
HRC	55, 50, 45, 40, 35, 30, 25, 20	"	"	"	"

Charpy V-Notch Test Blocks

In compliance with JIS B7740-1990
Feature: Extremely small irregularity in Charpy absorption energy (CV: 3% or less)
Material: SNCM439.Q.T (complying with JIS test piece No. 4)
NK verification provided.



Energy Level
Type A approx. 30J
Type B approx. 100J
Type C approx. 160J
5 pieces each energy levels per 1set

Standard Microstructure

Standard pieces of metal material microstructures, and detailed explanations with photographs and CD-ROM.



- Group 1. Carbon steel, Cast iron
- Group 2. Alloy tool steel, High-speed tool steel
- Group 3. Structural alloy steel, steel for special use
- Group 4. Non-ferrous alloys
- Group 5. Metallic surface, modifications
- Group 6. Abnormal Structure (23types)
- Group 7. Abnormal Structure (23types)

● A. G. S. ETCHING SET

Conforms to JIS G 0551
Etching set for revealing the prior Austenite Grainboundaries of Steels for structural use in machines.

Hardnester

Standard File for Scratch Hardness

PAT.NO.196592

Manual with CD-ROM

To measure metal hardness of various parts from steel material to thin layers simply and sensitively.

Hardnester	Rockwell "C" 8 Hardness Set	Vickers 8 Hardness Set	Vickers 16 Hardness Set
No. I	HR C 20	HV 200	HV 200-250
No. II	HR C 30	HV 300	HV 300-350
No. III	HR C 40	HV 400	HV 400-450
No. IV	HR C 50	HV 500	HV 500-550
No. V	HR C 55	HV 600	HV 600-650
No. VI	HR C 60	HV 700	HV 700-750
No. VII	HR C 65	HV 800	HV 800-850
No. VIII	HR C 67	HV 900	HV 900-950

with a spare contact measure for each hardness

■ Standard Piece for Spark Test

Manual with CD-ROM

Based on Japanese Industrial standard JIS G 0566-1980
Clearly identifies the species of JIS iron and steel materials.
Designed for educational or field use for machine assembly and heat treatment processes.



YAMAMOTO SCIENTIFIC TOOL LABORATORY CO.,LTD.

2-15-4, Sakae-cho, Funabashi-shi, Chiba Pref., 273-0018 Japan
Tel.+81-47-431-7451 Fax.+81-47-432-8592